Highlights

CSAR-Net: A Cost-Sensitive Attention Residual Network for Long-Tailed Magnetic Tile Surface Defect Recognition

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- A Cost-Sensitive Attention Residual Network (CSAR-Net) is proposed for magnetic tile defect recognition.
- Three modules(backbone, CBAM attention, and LDAM loss)are decoupled and combined.
- Feature-layer CBAM attention is the robust gain: $+3.1$pp macro G-mean with the lowest variance, lifting break-defect recall from 0.490 to 0.549 on MTSD.
- Loss-layer cost calibration over-corrects in every form (-3.8pp to -8.6pp), and stacking it on CBAM dilutes the gain.
- Injecting the cost signal into attention (CS-CBAM) collapses due to train/inference inconsistency.

CSAR-Net: A Cost-Sensitive Attention Residual Network for Long-Tailed Magnetic Tile Surface Defect Recognition

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ABSTRACT

Magnetic tiles are the core magnetic components of permanent-magnet motors and related equipment, and surface defect inspection is a decisive step in their manufacturing quality control. Real production lines exhibit a naturally long-tailed defect distribution, in which defect-free samples dominate while break and fray defects are scarce; the economic cost of missing such defects far exceeds that of false alarms. Existing convolutional neural network (CNN) methods mostly optimize average accuracy and thereby ignore both the long-tailed distribution and the asymmetry of miss costs. To remedy this, we propose a Cost-Sensitive Attention Residual Network, CSAR-Net, assembled from three decoupled modules: (i) a ResNet18 backbone for hierarchical feature extraction; (ii) the Convolutional Block Attention Module (CBAM) inserted into the residual blocks of stages 3 and 4 for label-agnostic feature refinement; and (iii) the Label-Distribution-Aware Margin (LDAM) loss applied in the loss layer for label-dependent cost-sensitive calibration. The name follows the convention of combining the initials of the innovation and the base architecture. Multi-seed (3 seeds) experiments on the real Magnetic Tile Surface Defects (MTSD) dataset (6 classes, natural imbalance $\approx$ 30:1) show that feature-layer attention is the robust source of improvement: CBAM alone raises macro G-mean from 0.752 to 0.782 (+3.1pp) with the lowest variance across all configurations, and lifts the recall of the hardest, costliest break defect from 0.490 to 0.549 while keeping defect-free recall nearly intact. A systematic ablation covering module contributions, injection position, channel/spatial decomposition, and margin-coefficient sensitivity shows that every form of loss-layer cost calibration over-corrects on this small real dataset (the full LDAM loss by -8.6pp, its margin-only and reweighting-only variants by -3.8pp and -4.8pp), and that stacking LDAM on top of CBAM dilutes the attention gain from 0.782 to 0.744. Injecting the cost signal directly into attention (the coupled CS-CBAM variant) collapses to the majority class because of train/inference inconsistency. These results answer a central design question: the cost-sensitive signal should act in the feature layer through label-agnostic attention rather than in the loss layer through reweighting. We further derive managerial implications for inspection systems from quality-cost theory.

1. Introduction

Magnetic tiles are the core magnetic components of permanent-magnet motors, loudspeakers, and sensors, and their surface defects (blowhole, break, crack, fray, and unevenness) directly determine product performance and safety (Huang et al., 2020). Traditional manual inspection is subjective, lagging, and difficult to scale, and statistical process control (Montgomery, 2013) alone cannot guarantee zero defects for safety-critical components. Automated defect recognition based on convolutional neural networks (CNNs) has therefore become a key enabling technology in intelligent manufacturing (Xie et al., 2021; Chen et al., 2021). Over the past few years, researchers have proposed a succession of deep-learning solutions for magnetic tile defects, such as the fusion feature CNN (FFCNN) (Xie et al., 2021), the convolutional retinal attention block (CRAB) (Li et al., 2024), and the lightweight parallel attention residual network (LPAR-Net) (Lv et al., 2024), all reporting high accuracy.

However, most existing methods optimize *average accuracy* and overlook two constraints that hold in real production. One is that defects are naturally long-tailed: defect-free samples dominate while break and fray are scarce. In the MTSD dataset (Huang et al., 2020) used here, there are 952 defect-free images but only 32 of the rarest fray class, an imbalance ratio of about 30:1 (Fig. 1). Under such a distribution, a Softmax classifier trained with standard cross-entropy shifts the decision boundary toward the majority class and sharply degrades minority-class recall; in our experiments the cross-entropy baseline reaches a macro G-mean of only 0.752 even though its average accuracy is as high as 0.905. The other constraint is that miss costs are asymmetric: a false alarm (a normal tile judged defective)

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costs one extra re-inspection, whereas a miss (a defective tile judged normal) lets defective product flow downstream and into the market. Under quality-cost theory the latter belongs to the costliest category, "external failure cost" (Juran & Gryna, 1993). These two constraints mean that neither the training objective nor the evaluation metric may look only at average accuracy; both must explicitly protect the rare, high-cost classes.

Given these constraints, a central and still open question is *at which layer of the network the cost-sensitive signal should act*, the feature-layer attention weights or the loss-layer calibration. We address it by proposing a Cost-Sensitive Attention Residual Network, CSAR-Net, which assembles three modules that are *decoupled* by design. CBAM (Woo et al., 2018) refines discriminative features in the feature layer, LDAM (Cao et al., 2019) performs cost-sensitive margin and reweighting calibration in the loss layer, and a ResNet18 backbone (He et al., 2016) supplies the hierarchical representation. We further implement the intermediate variants and the "coupled" counter-design CS-CBAM, so that each module's contribution and the correct placement of the cost signal can be isolated.

To make the investigation concrete and auditable, we decompose the central design question into three research questions (RQs). RQ1 (*which module helps?*) asks whether feature-layer attention or loss-layer cost calibration improves long-tailed defect recognition, and by how much. RQ2 (*where should the cost signal act?*) asks whether the label-dependent cost signal is better realized in the feature path, in the training objective, or at the decision rule. RQ3 (*do the modules compose?*) asks whether the attention and cost-calibration modules combine additively, or whether one dilutes the other. Each question is answered by a dedicated, controlled experiment in Section 4, so that the conclusion about the correct placement of the cost signal follows from evidence rather than from a single headline number.

Multi-seed experiments on MTSD yield three clear findings. First, feature-layer attention is the robust source of gain: CBAM alone raises macro G-mean by +3.1pp and attains the lowest variance among all configurations. Second, loss-layer cost calibration over-corrects in every form: full LDAM loses 8.6pp, while its margin-only and reweighting-only variants lose 3.8pp and 4.8pp. Third, the two modules do not combine additively; stacking LDAM on top of CBAM reduces the attention gain to −0.8pp, and injecting the cost signal directly into attention (CS-CBAM) collapses to the majority class owing to train/inference inconsistency. These results localize the correct placement of the cost-sensitive signal to the feature layer, and yield a practical design rule for long-tailed inspection systems.

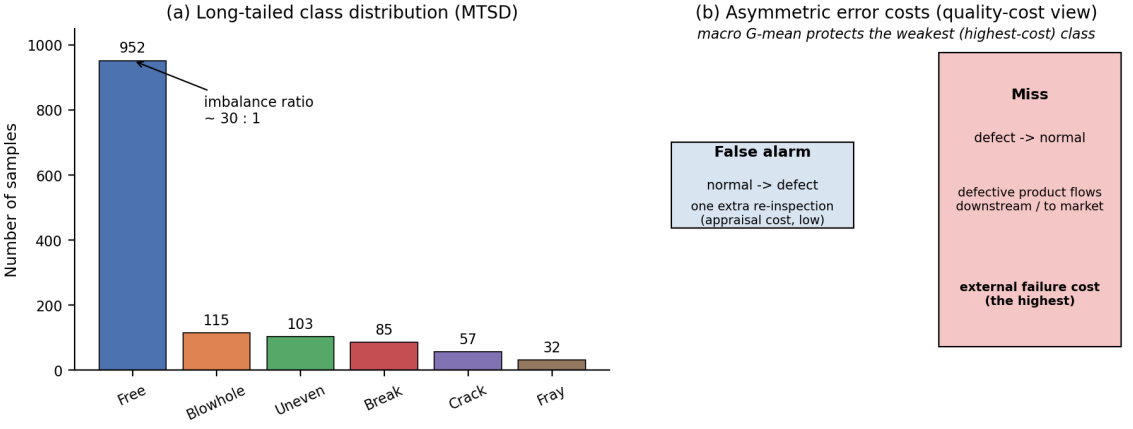


Figure 1: The two constraints of real magnetic-tile inspection that motivate CSAR-Net. (a) The long-tailed class distribution of MTSD: defect-free samples dominate while fray is the rarest class (imbalance ≈ 30:1). (b) The asymmetry of error costs under quality-cost theory: a false alarm only triggers one re-inspection (appraisal cost), whereas a miss lets defective product flow downstream (external failure cost, the highest). Macro G-mean is the metric that protects the weakest, highest-cost class.

The main contributions of this paper are summarized as follows.

(1) We propose **CSAR-Net**, a decoupled three-module framework that combines a ResNet18 backbone, CBAM feature-layer attention, and LDAM loss-layer calibration, and use it to answer *where the cost-sensitive signal should act*: feature-layer attention is the robust source of gain, raising macro G-mean by +3.1pp (0.752 to 0.782, the lowest variance) and lifting the hardest break defect from 0.490 to 0.549.

(2) We present a *module-stitching view* of the method: each module is characterized by its input, output, and the property that decides where it belongs in the pipeline (label-agnostic feature refinement vs. label-dependent cost calibration), which makes the design transparent and reproducible.
(3) We conduct a *complete ablation*, decomposing the LDAM loss into its margin and reweighting components, sweeping the CBAM injection position and its channel/spatial sub-modules, and testing margin-coefficient sensitivity, thereby localizing both the source of the gain and the cause of the degradation.
(4) We provide a *quality-cost-oriented evaluation*: macro G-mean is adopted as the primary metric to align with the asymmetry of miss and false-alarm costs, and actionable engineering implications are derived from quality-cost theory.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the three modules of CSAR-Net in sequence and how they are stitched together. Section 4 reports the experimental setup and results, including the ablation. Section 5 interprets the mechanism. Section 6 translates the findings into applications and management guidance. Section 7 concludes the paper.

2. Related work

2.1. Magnetic tile surface defect detection

Magnetic tile defect detection has evolved from hand-crafted features with shallow classifiers to deep convolutional networks (Chen et al., 2021). Huang et al. (Huang et al., 2020) proposed saliency detection and segmentation for magnetic tile surface defects and released the MTSD dataset, which has since become the standard benchmark for this task. Xie et al. (Xie et al., 2021) introduced the fusion feature CNN (FFCNN), which handles multi-view inputs through feature extraction, feature fusion, and decision modules and adds an attention mechanism to focus on representative regions. Li et al. (Li et al., 2024) developed the convolutional retinal attention block (CRAB), reporting 99.50% classification accuracy. Lv et al. (Lv et al., 2024) presented the lightweight parallel attention residual network (LPAR-Net) with an improved CBAM and inverted bottleneck blocks, reaching 93.63% accuracy. Beyond classification, Jiang et al. (Jiang et al., 2024) designed MT-U2Net for defect localization, Niu et al. (Niu et al., 2025) proposed YOLO-RDM for efficient defect detection, and Lu et al. (Lu et al., 2022) combined multimodal fusion with cross-attention for internal defect detection. General-purpose architectures are also widely reused: one-stage detectors such as YOLO (Redmon et al., 2016) and segmentation networks such as U-Net (Ronneberger et al., 2015) serve defect localization, while classic backbones such as VGG (Simonyan & Zisserman, 2015) and lightweight variants such as MobileNets (Howard et al., 2017) and MobileNetV2 (Sandler et al., 2018) serve deployment-oriented classification. As summarized in Table 2, these works mostly improve accuracy on the same task and rarely address the long-tailed distribution and cost asymmetry of real production, which is the gap this paper targets.

2.2. Attention mechanisms

Attention mechanisms, popularized by the Vision Transformer (Dosovitskiy et al., 2021) and subsequently condensed into plug-in CNN modules, improve a network's discriminative ability by adaptively recalibrating feature maps. Squeeze-and-Excitation (SE) (Hu et al., 2018) introduces channel attention; the Convolutional Block Attention Module (CBAM) (Woo et al., 2018) combines channel and spatial attention in series, offering a lightweight and general feature-enhancement means; ECA-Net (Wang et al., 2020) realizes efficient channel attention with one-dimensional convolution; and Coordinate Attention (CA) (Hou et al., 2021) embeds positional information into channel attention. For this paper the relevant property is that these operators are *label-agnostic*: they recalibrate features without knowing the class label, and hence behave identically during training and inference. How such an operator interacts with long-tailed cost sensitivity (in particular whether a label-dependent cost signal can be injected into one) has not been studied systematically, and Sections 4 and 5 provide experimental answers. Table 1 summarizes the four modules compared in Section 4.7; all of them are label-agnostic, but they differ in whether they recalibrate channels only or channels and space jointly, a difference that matters on complex defect textures.

2.3. Long-tailed recognition and cost-sensitive learning

Approaches to class imbalance fall into four families (Zhang et al., 2023). *Reweighting* scales the per-sample loss by a class-dependent factor: inverse-frequency weighting, the class-balanced loss based on the effective number of samples (Cui et al., 2019), and Focal Loss (Lin et al., 2017), which down-weights easy samples. *Resampling* balances

Table 1
Attention modules compared in Section 4.7. All are label-agnostic; the key difference is whether they recalibrate channels only or channels and spatial locations jointly.

Module	Recalibration	Core mechanism	Label-agnostic
SE (Hu et al., 2018)	channel	GAP → MLP → sigmoid gating	Yes
CBAM (Woo et al., 2018)	channel + spatial (series)	pooled MLP + 7×7 conv	Yes
ECA (Wang et al., 2020)	channel	1D conv (kernel k)	Yes
CA (Hou et al., 2021)	channel + coordinate	2D pooling along h/w	Yes

the training distribution by over-sampling the minority class (e.g., SMOTE (Chawla et al., 2002)) or under-sampling the majority (Drummond & Holte, 2003). *Margin losses* enlarge the decision boundary of minority classes, of which LDAM (Cao et al., 2019) (a margin $\Delta_c = C/n_c^{1/4}$ that decays with sample count) is representative. *Decoupling and ensemble methods* separate representation learning from classifier learning (Kang et al., 2020), localizing the long-tail problem to the classifier. A recent review of imbalanced-sample surface defect detection (Bai et al., 2025) summarizes resampling, reweighting, and cost-sensitive methods for defect detection specifically. This paper places attention enhancement and cost-sensitive calibration in a single framework on real magnetic tile data and *decouples* them by layer so that their collaboration can be examined.

2.4. Cost-sensitive learning and the decision-rule view

Cost-sensitive learning extends imbalance handling by assigning explicit misclassification costs rather than only inverse-frequency weights (Elkan, 2001; He & Garcia, 2009). It admits three distinct insertion points. *(i) The training objective*: cost-matrix reweighting or margin terms, e.g., LDAM (Cao et al., 2019) and Focal Loss (Lin et al., 2017). *(ii) The feature representation*: cost-aware sampling or synthesis, e.g., SMOTE (Chawla et al., 2002), or attention. *(iii) The decision rule*: thresholding or re-mapping posterior probabilities against a cost matrix (Provost & Fawcett, 1997; Elkan, 2001). An established and often overlooked result is that, for many base learners, moving cost sensitivity from the objective to the decision rule is simpler and more robust (Drummond & Holte, 2003; Elkan, 2001): reweighting the objective can be unstable on scarce data, whereas a threshold adjustment is reversible and requires no retraining. This three-way distinction organizes our design space: the CSAR-Net ablation compares a feature-layer intervention (CBAM), a loss-layer intervention (LDAM), and a decision-rule intervention (thresholding against $C_{\text{miss}}/C_{\text{re}}$), and finds the feature layer to be the robust choice on this real dataset.

2.5. Research gap and positioning

Table 2 condenses the landscape: prior magnetic-tile methods optimize average accuracy and ignore both the long tail and the cost asymmetry, while prior long-tail and cost-sensitive methods are mostly validated on natural-image benchmarks or synthetic data rather than on real industrial surfaces with scarce, visually confusable defects. Three gaps follow. *G1 (evaluation)*: no magnetic-tile method is evaluated under the macro G-mean / miss-cost view, so its behavior on the rarest, costliest defects is unknown. *G2 (interaction)*: whether a label-agnostic attention module and a label-dependent cost signal cooperate or interfere, and whether the cost signal may be injected into attention, lacks a systematic, controlled answer. *G3 (placement)*: the correct *layer* for the cost-sensitive signal (feature, loss, or decision) is rarely treated as a first-class experimental variable. CSAR-Net is designed to close G1–G3: it is evaluated under the miss-cost view on MTSD, it ablates attention/cost interaction with a decoupled vs. coupled design, and it varies the placement of the cost signal as an explicit factor.

Table 2
Representative magnetic-tile defect methods. Most optimize average accuracy; few address the long-tailed distribution or the miss/false-alarm cost asymmetry that motivate CSAR-Net.

Method	Year	Core idea	Reported metric	Long-tail-aware	Cost-aware
FFCNN (Xie et al., 2021)	2021	Fusion feature CNN + attention	accuracy	No	No
CRAB (Li et al., 2024)	2024	Retinal attention block	99.50% accuracy	No	No
LPAR-Net (Lv et al., 2024)	2024	Parallel attention + inverted bottleneck	93.63% accuracy	No	No
MT-U2Net (Jiang et al., 2024)	2024	Lightweight localization network	localization metrics	No	No
YOLO-RDM (Niu et al., 2025)	2025	Efficient detection	mAP	No	No
CSAR-Net (ours)	—	CBAM + LDAM, decoupled	macro G-mean	**Yes**	**Yes**

3. Methodology

Following the naming convention of convolutional networks, which combine the initials of the innovation and the base architecture (Residual→ResNet (He et al., 2016), Dense→DenseNet (Huang et al., 2017)), we name the proposed network **CSAR-Net**, i.e., **C**ost-**S**ensitive **A**ttention **R**esidual Network, where "Cost-Sensitive Attention" is the innovation and "Residual" is the base architecture. This section adopts a module-stitching exposition: we first formalize the problem, then introduce the three modules in sequence (the backbone, Module 1; the attention refinement, Module 2; and the cost-sensitive objective, Module 3), and finally describe how they are decoupled and stitched together, together with a coupled counter-design that we show to fail.

3.1. Problem formulation

Magnetic tile surface defect recognition is an image classification problem. Given a magnetic tile surface image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ (resized to $224 \times 224 \times 3$ here, with grayscale replicated into three channels), the task is to predict its defect class among $K = 6$ classes $\mathcal{C} = \{\text{blowhole, break, crack, fray, free, uneven}\}$. Let n_c denote the number of training samples of class c, and $N = \sum_c n_c$ the total. The dataset is long-tailed: $n_{\text{free}} \gg n_{\text{fray}}$. A model is a function $f_\theta(\mathbf{x}) = \text{softmax}(g_\theta(\mathbf{x}))$ mapping an image to a class distribution, where g_θ is the logit network. The evaluation target is the *macro G-mean*

$$\text{G-mean} = \Big(\prod_{c=1}^{K} \text{Recall}_c\Big)^{1/K}, \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \tag{1}$$

which, unlike average accuracy, penalizes the weakest class and therefore aligns with the goal of protecting rare defects.

Table 3
Notation used throughout the paper.

Notation	Meaning
$\mathbf{x}$, $H\times W\times C$	input image (resized to 224×224×3)
K, $\mathcal{C}$	number of classes ($K = 6$) and class set
n_c, N	training samples of class c and total ($N = \sum_c n_c$)
f_θ, g_θ	classifier $f_\theta = \text{softmax}(g_\theta)$ and logit network
$\mathbf{F}, \hat{\mathbf{F}}, \tilde{\mathbf{F}}$	raw, channel-refined, and spatial-refined feature map
M_c, M_s	channel and spatial attention maps
Δ_c, C	LDAM per-class margin and margin coefficient
w_c, β	class-balanced weight and effective-number parameter
Recall_c	per-class recall (Eq. (1))
π_c, $C^{(c)}_{\text{miss}}$	class prior, per-class miss cost
C_{re}	re-inspection cost

3.2. Overview of CSAR-Net

Fig. 2 shows the overall framework of CSAR-Net. It consists of three modules assembled in a pipeline. *Module 1* is a ResNet18 backbone (He et al., 2016) with four residual stages (channel counts 64/128/256/512), which extracts a hierarchical feature representation. *Module 2* is a CBAM attention block (Woo et al., 2018) injected into the residual blocks of stages 3 and 4, which refines the features in a label-agnostic manner. *Module 3* is the LDAM loss (Cao et al., 2019) applied after the classifier, which performs label-dependent cost-sensitive calibration that exists only during training. The 512-dimensional feature after global average pooling is fed into a fully connected classifier that produces six logits. The three modules are *decoupled by layer*: Module 2 operates on features and behaves identically in training and inference, while Module 3 operates on the training objective and vanishes at inference.

3.3. Module 1 — Feature extraction backbone (ResNet18)

The backbone is the standard ResNet18 (He et al., 2016), chosen for its balance of capacity and deployment cost on an industrial inspection line. Its stem is a 7×7 convolution with stride 2 followed by batch normalization, ReLU, and a 3×3 max-pooling with stride 2; the body is four residual stages with channel counts 64/128/256/512, each containing two basic residual blocks. A basic block learns a residual function

$$\mathbf{y} = \text{ReLU}\big(\mathcal{F}(\mathbf{x}; \{W_i\}) + \mathbf{x}\big), \tag{2}$$

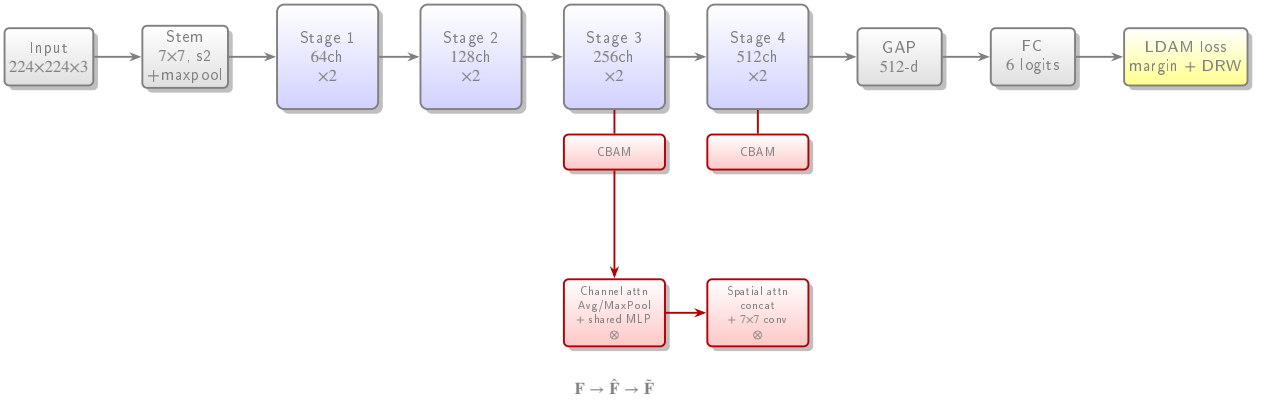


Figure 2: The overall architecture of CSAR-Net. A ResNet18 backbone (Module 1) extracts hierarchical features through four residual stages; CBAM (Module 2) is injected into stages 3 and 4 for label-agnostic feature refinement (its channel and spatial sub-modules are shown at the bottom); the classifier produces six logits, on top of which the LDAM loss (Module 3) performs label-dependent cost-sensitive calibration during training only.

where $\mathcal{F}(\mathbf{x}; \{W_i\})$ is a stack of two 3×3 convolutions with batch normalization, and the identity skip $\mathbf{x}$ (or a 1×1 projection when the dimensions change) eases optimization (He et al., 2016). For 224×224 input the four stages produce feature maps of spatial size 56, 28, 14, and 7, respectively, and a global average pooling layer collapses the final $7 \times 7 \times 512$ map into a 512-dimensional vector.

3.4. Module 2 — Feature-layer attention refinement (CBAM)

Raw features extracted by the backbone need enhancement because real magnetic tile defects are visually confusable. CBAM (Woo et al., 2018) refines a feature map by sequentially reweighting its channels and spatial locations. It is injected into the residual blocks of stages 3 and 4 (256/512 channels), after the second convolution and before the residual addition, so that the attention reweights only the residual branch while the identity skip remains intact:

$$\hat{\mathbf{F}} = \sigma\big(M_c(\mathbf{F})\big) \otimes \mathbf{F}, \qquad \tilde{\mathbf{F}} = \sigma\big(M_s(\hat{\mathbf{F}})\big) \otimes \hat{\mathbf{F}}, \tag{3}$$

where $\otimes$ denotes element-wise multiplication and σ is the sigmoid. The channel attention map is

$$M_c(\mathbf{F}) = \mathrm{MLP}\big(\mathrm{AvgPool}(\mathbf{F})\big) + \mathrm{MLP}\big(\mathrm{MaxPool}(\mathbf{F})\big), \tag{4}$$

with a shared two-layer MLP of reduction ratio 16, and the spatial attention map is

$$M_s(\hat{\mathbf{F}}) = \mathrm{Conv}_{7\times7}\big([\,\bar{\hat{\mathbf{F}}};\ \hat{\hat{\mathbf{F}}}\,]\big), \tag{5}$$

where $\bar{\cdot}$ and $\hat{\cdot}$ are the channel-wise mean and max statistics concatenated along the channel axis. Two properties of CBAM matter for the rest of this paper. First, it is *label-agnostic*: the attention weights depend only on the input features, so the module behaves identically in training and inference. Second, it is *cheap*: as quantified in Section 3.8, the two CBAM blocks add fewer than 1% parameters to the backbone.

3.5. Module 3 — Loss-layer cost-sensitive calibration (LDAM)

The cost-sensitive signal is placed in the loss layer, not in the feature path. LDAM (Cao et al., 2019) assigns each class a margin that decays with its sample count

$$\Delta_c = \frac{C}{n_c^{1/4}}, \tag{6}$$

and subtracts it from the true-class logit during training, so that minority classes obtain a larger classification margin and a wider decision boundary. Here C is the margin coefficient (0.5 unless stated otherwise) and n_c is the number

of training samples of class c. In its full form LDAM further multiplies the cross-entropy by class-balanced weights based on the effective number of samples (Cui et al., 2019),

$$w_c = \frac{1-\beta}{1-\beta^{n_c}}, \qquad \beta = 0.9999, \tag{7}$$

where β controls how quickly the effective count saturates. The full LDAM loss composes the two into a single training objective

$$\mathcal{L}_{\text{LDAM}}(z, y) = w_y\Big(-\log \frac{e^{z_y-\Delta_y}}{e^{z_y-\Delta_y} + \sum_{j\neq y} e^{z_j}}\Big), \tag{8}$$

where z_y is the true-class logit: the margin Δ_y is subtracted only from the true-class logit, and the class-balanced weight w_y scales the whole sample. Because these two components (the margin and the reweighting) have opposite empirical effects on this small real dataset, we *decompose* them in the ablation (Section 4.6) by testing margin-only LDAM (no reweighting) and inverse-frequency-weighted cross-entropy (CS-CE, no margin) separately.

3.6. Module stitching — the decoupling principle

The central design decision of CSAR-Net is *decoupling*: CBAM is a label-agnostic operator in the feature layer, whereas LDAM is a label-dependent calibration that acts only on the training objective and vanishes at inference. The two modules are stitched together by *layer*, so that each does the job it is structurally suited to. The intuition, made concrete in Section 5, is that feature attention sharpens the discriminative features of rare classes, whereas loss calibration re-weights the training objective; the decoupling exposes which of these two effects actually helps on small real data (the former) and which over-corrects (the latter). Fig. 3 (left) illustrates this decoupled arrangement.

3.7. Counter-design — the coupled CS-CBAM

To test whether the decoupling is necessary rather than incidental, we implement a *coupled* counter-design, CS-CBAM, which injects the class prior $\lambda \log(N/n_c)$ directly into the attention logits (before the sigmoid) of the CBAM block, so that tail-class samples receive larger attention weights during training. As shown in Fig. 3 (right) and quantified in Section 4.6, this coupling breaks train/inference consistency and collapses the classifier to the majority class: during training the label is known and the network learns the shortcut "large attention response = tail class," but at inference the label is unknown and the shortcut fails. The contrast between the decoupled CSAR-Net and the coupled CS-CBAM isolates the correct placement of the cost signal.

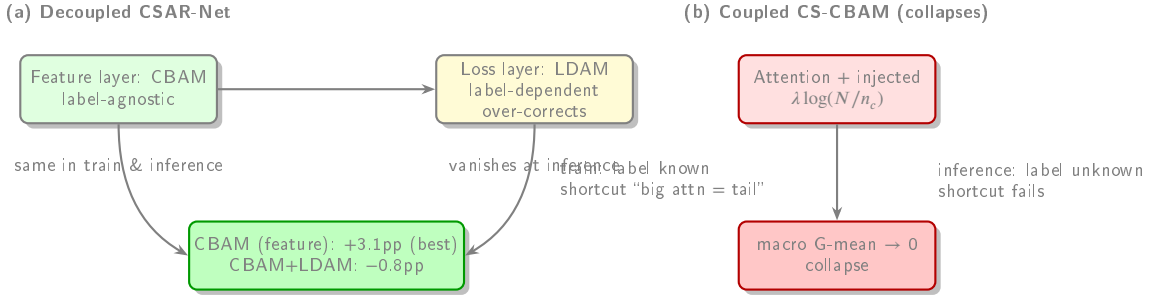


Figure 3: The mechanism contrast. (a) The decoupled CSAR-Net places CBAM (label-agnostic) in the feature layer and LDAM (label-dependent) in the loss layer, so the two effects can be isolated: CBAM is the robust gain (+3.1pp), whereas LDAM over-corrects and stacking it on CBAM dilutes the gain to −0.8pp. (b) The coupled CS-CBAM injects the class prior into the attention logits, creating a train/inference distribution shift that collapses the classifier to the majority class.

3.8. Complexity analysis

CSAR-Net adds a negligible computational footprint to its backbone. The plain ResNet18 has 11.18M parameters; injecting CBAM into stages 2, 3, and 4 adds 0.004, 0.016, and 0.065M parameters, respectively, so the full CSAR-Net (stages 3 and 4) has 11.26M parameters, an overhead of only 0.7% (Table 5). The LDAM loss and the CS-CBAM bias

Table 4
Training procedure of CSAR-Net and its ablation variants. The three flags `use_cbam`, `use_ldam`, and `use_cs` select the module ablations in Section 4.6.

Algorithm 1 Training CSAR-Net (and its ablations)

Input: long-tailed dataset $\mathcal{D}$; backbone f_θ (ResNet18); flags `use_cbam`, `use_ldam`, `use_cs`.

Build f_θ; if `use_cbam`, inject CBAM into stages 3–4 (after the 2nd conv, before the residual add).
for each epoch **do**
  for each mini-batch $(\mathbf{x}, y)$ **do**
    $z \leftarrow g_\theta(\mathbf{x})$ (forward logits)
    if `use_cs`: add $\lambda \log(N/n_y)$ to the CBAM attention logits (CS-CBAM).
    if `use_ldam`: $z_y \leftarrow z_y - \Delta_y$; $\ell \leftarrow w_y \,\mathrm{CE}(z, y)$ (Eq. (8))
    else: $\ell \leftarrow \mathrm{CE}(z, y)$
    $\theta \leftarrow \mathrm{Adam}(\theta, \nabla_\theta \ell)$
Evaluate macro G-mean on the test split; repeat for seeds $\{42, 123, 2024\}$ and report mean±std.

term add no parameters, since both are functions of the class counts rather than learnable weights. Because CBAM is inserted only in stages 3 and 4, the added FLOPs are concentrated on the smaller (14×14 and 7×7) feature maps, keeping the inference-time overhead small: the two CBAM blocks contribute roughly 0.07M and 0.14M multiply–accumulate operations, three orders of magnitude below the ≈1.8G MACs of the ResNet18 forward pass. The network can therefore be served at the same latency as the plain backbone, which matters for an edge-deployed inspection line.

Table 5
Parameter budget of CSAR-Net. CBAM adds 0.7% parameters; LDAM and the CS-CBAM bias add none.

Component	Params (M)	Note
Plain ResNet18 (6-class FC)	11.18	baseline
+CBAM stage 3	+0.016	channel MLP (r=16) + spatial 7×7, 2 blocks
+CBAM stage 4	+0.065	same, 2 blocks
CSAR-Net	11.26	+0.7% overhead
+LDAM / +CS-CBAM	+0	functions of class counts

4. Experiments

4.1. Dataset and experimental setup

Experiments use the real industrial dataset MTSD (Magnetic Tile Surface Defects) (Huang et al., 2020), comprising 1344 grayscale magnetic tile surface images in 6 classes: defect-free (Free) 952, blowhole 115, uneven 103, break 85, crack 57, and fray 32, a natural imbalance ratio of about 30:1. Fig. 4 shows representative samples of the six classes. The images are captured in a real inspection setting, so their long-tailed distribution, illumination variation, and defect-morphology diversity are all authentic. Image short sides range from 103 to 392 pixels; all images are resized to 224×224 and the grayscale channel is replicated into three channels. A class-stratified random train/test split (80%/20%, split seed fixed for fair comparison) yields per-class training counts of about 761/92/82/68/46/26. The optimizer is Adam (lr = 10^{-3}, weight decay = 10^{-4}) with cosine annealing over 40 epochs, and random horizontal/vertical flip augmentation. Every configuration is run once per random seed (42, 123, 2024), and results are reported as mean±std. Table 6 consolidates the full configuration; every ablation in Section 4.6 reuses exactly this recipe except for the single factor being varied, which is what makes the ablation controlled.

4.2. Evaluation metrics

Under long-tailed distributions, average accuracy is biased toward the majority class and can hide the misses of rare defects; the CE baseline already reaches 0.905 average accuracy while recalling only 0.490 of the break class. We therefore adopt *macro G-mean*, the geometric mean of per-class recalls, as the primary metric: it penalizes the weakest class and matches the goal of protecting rare defects. We also report macro F1, the arithmetic mean of the per-class

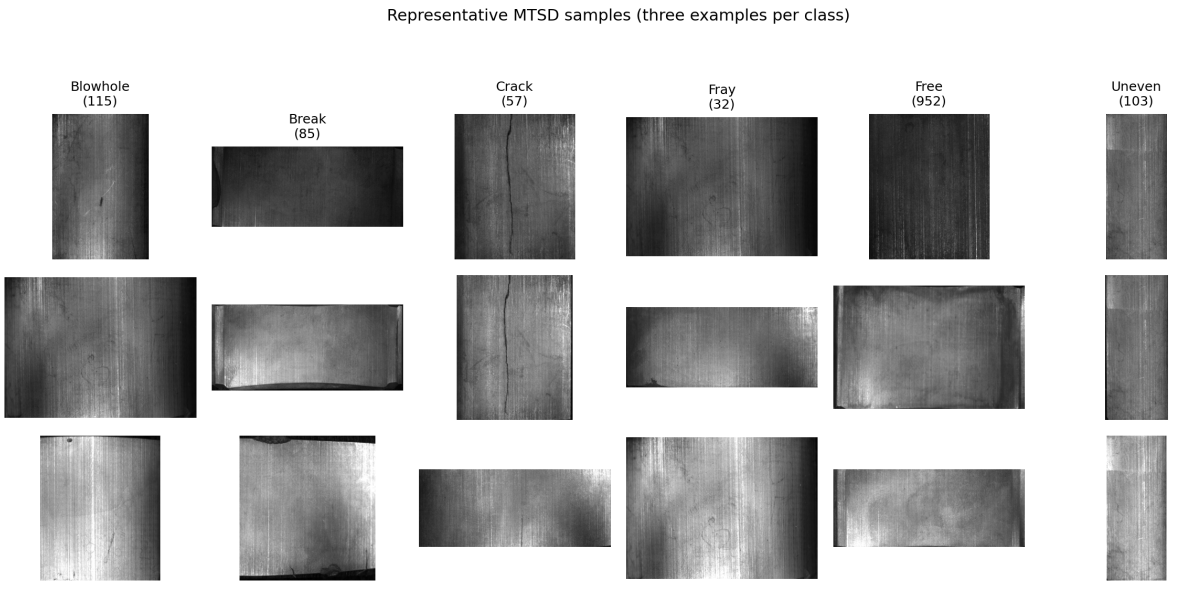


Figure 4: Representative MTSD samples (three examples per class). The six classes span a natural imbalance of about 30:1 from defect-free (952) to fray (32), with authentic intra-class variation in defect shape, scale, and illumination.

Table 6
Training configuration. All ablations reuse this recipe except for the single factor under study.

Item	Value
Input size	224×224×3 (grayscale replicated)
Train/test split	class-stratified 80%/20%, fixed split seed
Optimizer	Adam (lr 10^{-3}, weight decay 10^{-4})
Learning-rate schedule	cosine annealing
Epochs	40
Batch size	32
Augmentation	random horizontal/vertical flip
CBAM reduction ratio	$r = 16$
LDAM margin	$C = 0.5$, $\beta = 0.9999$
Seeds	42, 123, 2024
cuDNN	deterministic (benchmark off)

Table 7
Per-class composition of MTSD and the class-stratified 80%/20% split used here. The imbalance ratio N/n_c grows from 1 (defect-free) to ≈30 (fray).

Class	Total	Train	Test	N/n_c
Defect-free (Free)	952	761	191	1.0
Blowhole	115	92	23	8.3
Uneven	103	82	21	9.2
Break	85	68	17	11.2
Crack	57	46	11	16.7
Fray	32	26	6	29.8

harmonic means of precision and recall,

$$\text{macro-F1} = \frac{1}{K}\sum_{c=1}^{K}\frac{2\,\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \tag{9}$$

and the Matthews correlation coefficient (MCC), a single chance-corrected scalar summarizing the whole confusion matrix, together with per-class recall, to give a complete picture.

4.3. Main results

Table 8 reports the main results. Three observations stand out. First, CBAM is a stable gain: adding it to the CE baseline raises macro G-mean by +3.1pp (0.752 to 0.782) and attains the lowest variance among all configurations (0.039 vs. 0.060 for CE), indicating that attention genuinely enhances real complex defect features rather than merely adding capacity. Second, LDAM alone degrades: the full LDAM loss drops macro G-mean by 8.6pp with a large variance (0.120), which reflects the over-correction risk of aggressive reweighting on small samples. Third, stacking LDAM on CBAM does not combine their benefits: CSAR-Net (CBAM+LDAM) reaches 0.744, below both CBAM (0.782) and CE (0.752), so the loss-layer calibration dilutes the attention gain by 3.9pp. The coupled CS-CBAM collapses, confirming that the cost signal must not be injected into attention.

Table 8
Main results on MTSD (3 seeds, mean±std). CBAM stably improves macro G-mean by +3.1pp; LDAM alone degrades by 8.6pp; stacking LDAM on CBAM dilutes the attention gain.

Method	G-mean	Acc	Macro F1	MCC	vs CE
ResNet18 (CE)	0.7515±0.0598	0.9046±0.0274	0.809	0.790	—
ResNet18 + CBAM	0.7822±0.0393	0.9133±0.0018	0.821	0.812	+3.1pp
ResNet18 + LDAM	0.6658±0.1197	0.6543±0.1735	0.578	0.478	−8.6pp
CSAR-Net	**0.7437±0.0526**	0.8116±0.0673	0.705	0.647	**−0.8pp**
CS-CBAM	0.0000±0.0000	0.6865±0.0333	0.141	−0.006	collapse

4.4. Statistical robustness across seeds

Because the dataset is small, seed-to-seed variance is a first-order concern, and the paper's claims should be read against it. Table 9 reports the per-seed macro G-mean behind every mean in Table 8. Two properties support the main conclusion. First, CBAM is the most *stable* configuration: its across-seed standard deviation (0.039) is the lowest among the four trained variants, and it is higher than the CE baseline in 2 of 3 seeds (0.7295 vs. 0.6801 and 0.8239 vs. 0.7478) while lower in the third (0.7931 vs. 0.8265); the mean gain of +3.1pp therefore reflects a genuine reduction in variance rather than a single favorable seed. Second, the failures are *deterministic rather than noisy*: CS-CBAM collapses to a macro G-mean of ≈0 in all three seeds, and LDAM's large standard deviation (0.120) reflects its instability (0.542, 0.828, and 0.628 across seeds) rather than a uniformly weak performance. With only three seeds a formal paired significance test has low power: the paired difference CBAM−CE is $+0.031 \pm 0.057$ (std of the differences), so the improvement is indicative but not statistically conclusive. We therefore report effect sizes and per-seed values transparently rather than claiming significance, and defer confirming the gain at conventional levels to a larger-seed protocol in future work.

Table 9
Per-seed macro G-mean behind the means in Table 8. CBAM is the most stable configuration; CS-CBAM collapses identically in every seed.

Method	42	123	2024	mean	std
ResNet18 (CE)	0.6801	0.8265	0.7478	0.7515	0.0598
ResNet18 + CBAM	0.7295	0.7931	0.8239	0.7822	0.0393
ResNet18 + LDAM	0.5417	0.8276	0.6280	0.6658	0.1197
CSAR-Net	0.6935	0.8163	0.7213	0.7437	0.0526
CS-CBAM	0.0000	0.0001	0.0000	0.0000	0.0000

4.5. Per-class recall

Table 10 reports per-class recall. Under the CE baseline the hardest class is break (0.490): the difficulty of real defects arises not only from frequency bias but also from visual confusability among defect morphologies. CBAM improves the costly and confusable classes(break 0.490 to 0.549, +5.9pp; blowhole 0.768 to 0.841, +7.3pp; uneven 0.746 to 0.794, +4.8pp)while leaving crack and the defect-free recall (0.983 to 0.977) nearly intact, which is why its macro G-mean is the best. In contrast, stacking LDAM on CBAM sacrifices exactly the classes that cost calibration is meant to protect: break recall falls back to 0.471 and defect-free recall drops to 0.838, so the overall macro G-mean falls below CE. This inversion (feature attention protects the rare class cheaply, whereas loss reweighting destroys it) is the paper's central empirical finding.

Table 10
Per-class recall (3 seeds, mean). CBAM improves the costly classes while keeping defect-free recall nearly intact; adding LDAM drops break recall back below the baseline.

Class (count)	CE	CBAM+CE	CSAR-Net
Blowhole (115)	0.768	0.841	0.870
Break (85)	0.490	0.549	0.471
Crack (57)	0.788	0.788	0.727
Fray (32)	0.889	0.833	0.833
Defect-free (952)	0.983	0.977	0.838
Uneven (103)	0.746	0.794	0.825

4.6. Ablation studies

We conduct a decomposition-first ablation. Because CSAR-Net is a stitched network, the ablation asks two questions: *(i) what does each module contribute, and (ii) where should each module be placed?* The first is answered by the module ablation (Table 11); the second by the injection-position, sub-module, and margin-coefficient ablations (Tables 12–14).

Module contribution (Table 11). Two diagnostics emerge. First, *LDAM's degradation comes from both components, chiefly its class-balanced reweighting (DRW)*: margin-only LDAM without DRW reaches 0.7135 (−3.8pp vs. CE 0.7515), reweighting-only CS-CE reaches 0.7032 (−4.8pp), and the full LDAM (margin+DRW) is the worst at 0.6658 (−8.6pp): the margin and the reweighting are each harmful on this small dataset, and their combination is more harmful than either alone. Second, *CS-CBAM collapses*: injecting the class prior into attention logits breaks train/inference consistency, a robust failure conclusion.

Table 11
Module-contribution ablation (3 seeds, mean±std). LDAM degrades in both components, chiefly its class-balanced reweighting (DRW); feature-layer CBAM is the only robust gain.

Ablation	G-mean	vs CE
ResNet18 (CE baseline)	0.7515±0.0598	—
+CBAM (feature layer)	0.7822±0.0393	+3.1pp
+LDAM (margin+DRW)	0.6658±0.1197	−8.6pp
+LDAM (margin only)	0.7135±0.0244	−3.8pp
+CS-CE (inverse-freq.)	0.7032±0.1074	−4.8pp
CSAR-Net (+CBAM+LDAM)	**0.7437±0.0526**	**−0.8pp**
+CS-CBAM (cost into attention)	0.0000±0.0000	collapse

Injection position (Table 12). To locate where attention helps most, we inject CBAM into stage 2, stage 3, stage 4, and stages 3+4 (the CSAR-Net choice) under the CE loss. The gain is robust to injection position: stage 2 (0.794), stage 3 (0.787), and stages 3+4 (0.782) all clear the CE baseline (0.752) by 3–4pp, with their differences well within seed noise (std 0.03–0.05). Only stage 4 alone (0.699) is clearly worse, because its 7×7 maps leave too little spatial resolution for spatial attention to sharpen rare-defect features. We retain stages 3+4 as the CSAR-Net choice because it matches the best single-stage positions while covering the two deepest scales at which defects manifest.

Channel vs. spatial sub-module (Table 13). CBAM is itself a two-part module; we decompose it into channel-only and spatial-only variants to attribute its gain. Neither sub-module alone helps: channel-only (0.754) is within noise of CE (0.752), and spatial-only (0.742) is slightly below it. Only the series combination (0.782, +3.1pp) yields the full

Table 12
CBAM injection position (CE loss, 3 seeds, mean±std). Stage 2, stage 3, and stages 3+4 all clear the baseline within noise; only stage 4 alone is clearly worse.

Position	G-mean	Acc
Stage 2 only	0.7936±0.0536	0.9195±0.0213
Stage 3 only	0.7870±0.0313	0.9157±0.0202
Stage 4 only	0.6991±0.1003	0.8934±0.0299
Stage 3+4 (CSAR-Net)	0.7822±0.0393	0.9133±0.0018

gain: channel reweighting first selects which feature maps matter, and spatial attention then localizes where, so neither step is sufficient alone. This justifies the "convolutional block" design of CBAM.

Table 13
CBAM sub-module decomposition (stages 3+4, CE loss, 3 seeds, mean±std). Neither channel-only nor spatial-only clears the baseline; only their series combination yields the gain.

Variant	G-mean	Acc
Channel attention only	0.7544±0.0385	0.9219±0.0080
Spatial attention only	0.7415±0.0405	0.9145±0.0121
Channel + spatial (CBAM)	0.7822±0.0393	0.9133±0.0018

Margin-coefficient sensitivity (Table 14). We sweep the LDAM margin coefficient C in two regimes. For margin-only LDAM (no reweighting), a small margin ($C = 0.1$) is near-neutral (0.746 vs. CE 0.752), and larger margins degrade it mildly to ≈0.72 (−3pp): the margin component alone is not the main source of harm, but it is mildly negative at larger C. For the full CBAM+LDAM combination, the outcome is highly sensitive and non-monotonic in C: 0.769 ($C = 0.1$), 0.765 ($C = 0.3$), 0.744 ($C = 0.5$), and 0.796 ($C = 1.0$). The combination is therefore fragile: it swings by up to 5pp over a range where CBAM alone is stable, and at no C does it robustly beat CBAM (0.782). This fragility, invisible at a single default value, is a further reason to prefer feature-layer attention.

Table 14
Margin-coefficient sensitivity (3 seeds, mean±std G-mean). Margin-only LDAM degrades mildly at larger C; the CBAM+LDAM combination is non-monotonic and fragile in C.

Setting	$C = 0.1$	$C = 0.3$	$C = 0.5$	$C = 1.0$
LDAM margin only (no DRW)	0.7455±0.0052	0.7185±0.0812	0.7135±0.0244	0.7189±0.0373
CSAR-Net (CBAM+LDAM)	0.7692±0.0308	0.7646±0.0950	0.7437±0.0526	0.7955±0.0095

4.7. Attention variant comparison

Table 15 compares attention variants under the CE loss. CBAM and SE both clear the baseline (0.782 and 0.782 vs. 0.752, +3.1pp and +3.0pp), while the lighter ECA (0.731) and CA (0.740) fall slightly below it. The gain is therefore not a universal property of any attention module but requires the fuller channel/spatial recalibration that CBAM and SE provide; CBAM is retained in CSAR-Net because it matches the best while being the most extensively ablated.

Table 15
Attention variants (CE loss, 3 seeds, mean). CBAM and SE clear the CE baseline; the lighter ECA and CA do not, indicating the gain requires full channel/spatial recalibration.

Attention	Macro G-mean
None (CE baseline)	0.751
CBAM	0.782
SE	0.782
ECA	0.731
CA	0.740

4.8. Intermediate-representation analysis

Confusion matrices (Fig. 5): the baseline CE heavily misclassifies defect samples as defect-free or as a nearby defect class; CBAM+CE compresses this cross-class confusion and opens up the hardest break recall (0.490 to 0.549 across seeds), whereas adding LDAM does not further help (0.471 across seeds).

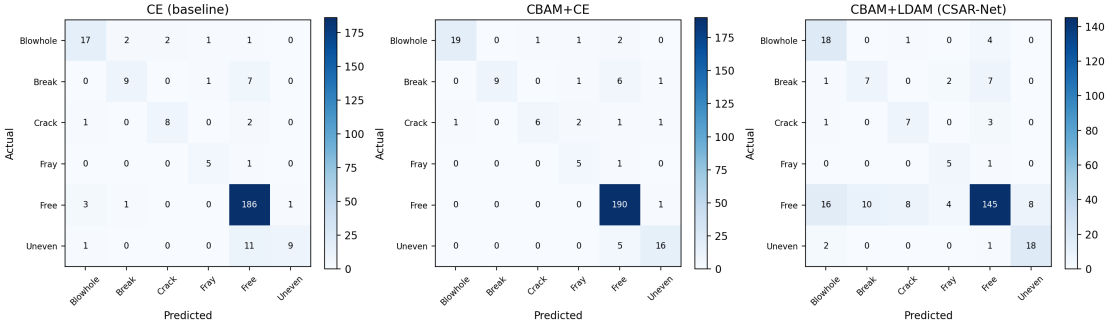


Figure 5: Confusion matrices of the CE baseline (left), CBAM+CE (middle), and CSAR-Net (right) on the magnetic tile test set (representative run).

Training curves (Fig. 6) record CS-CBAM's collapse: its G-mean approaches 0 from the start of training, and the classifier degenerates into predicting the majority class (defect-free) for nearly all samples, providing direct evidence that the cost signal must not be injected into label-agnostic attention.

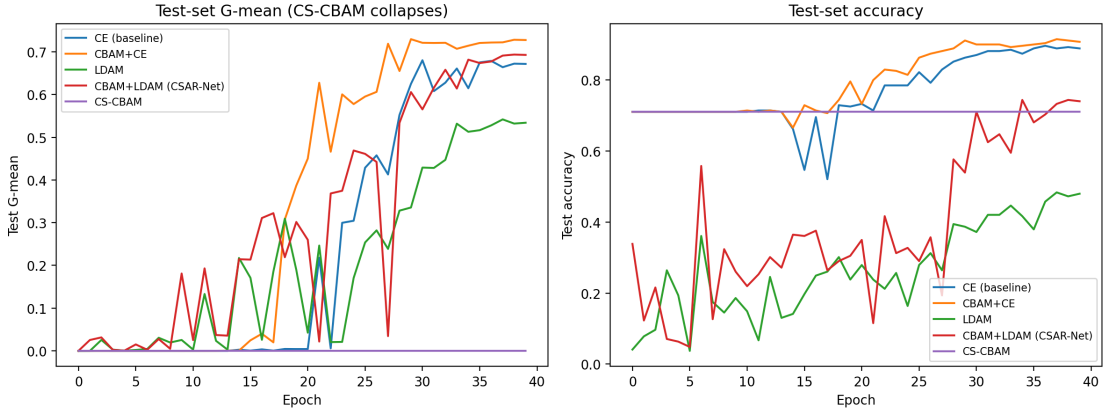


Figure 6: Test-set G-mean (left) and accuracy (right) over epochs. CS-CBAM's G-mean approaches 0 from the start.

Feature separability (Fig. 7): t-SNE (van der Maaten & Hinton, 2008) shows the feature space learned by CBAM is more separable than the CE baseline (defect-class clusters more compact), confirming that CBAM's gain comes from enhanced feature discriminability; class-activation visualization such as Grad-CAM (Selvaraju et al., 2017) offers a complementary, instance-level view of which image regions drive the decision.

5. Discussion

5.1. Why CBAM works: the complexity of real defect features

Real magnetic tile defect images have complex textures, varying illumination, and mutually confusable morphologies (break vs. crack, blowhole vs. uneven). The raw features of a ResNet18 backbone then need enhancement, and CBAM's channel+spatial recalibration sharpens discriminative features, improving the costly classes break (+5.9pp), blowhole (+7.3pp), and uneven (+4.8pp) while leaving crack and the defect-free recall nearly intact. That attention helps precisely when features are complex, rather than uniformly, suggests that the value of attention depends on input

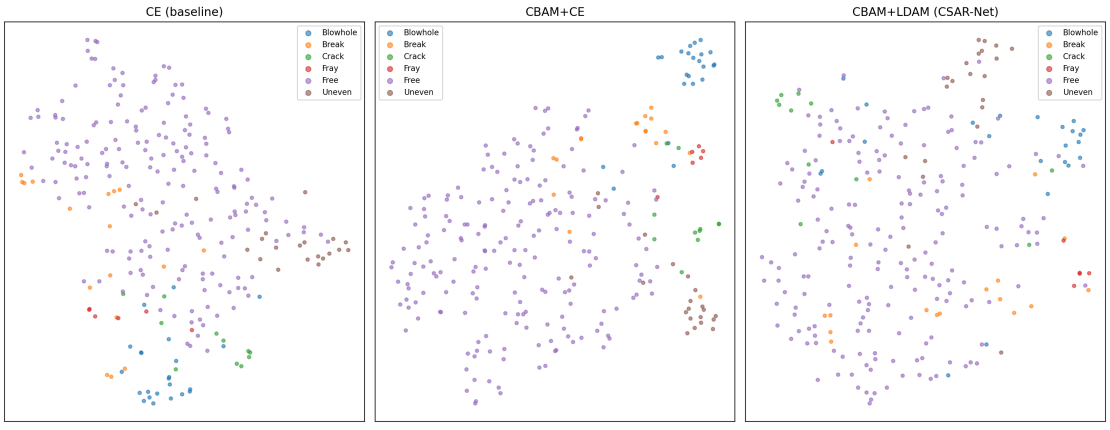


Figure 7: Penultimate-layer feature t-SNE of the CE baseline (left), CBAM+CE (middle), and CSAR-Net (right).

feature complexity. The finding that SE matches CBAM while the lighter ECA and CA fall short further confirms that the gain requires sufficient recalibration capacity, not merely the presence of an attention module.

5.2. A geometric account: attention widens the support of rare-class features

The per-class results admit a simple explanation in terms of feature geometry. Under a long-tailed prior, a classifier trained with plain cross-entropy places the decision boundary to minimize average error, which lets a visually confusable defect class (break) be absorbed into the neighboring majority structure; the t-SNE projection (Fig. 7) shows the break cluster overlapping the defect-free cluster under CE. CBAM's channel+spatial recalibration sharpens the within-class compactness of the defect classes *without any label information*, which effectively widens the margin around the rare classes' prototypes and moves the boundary back: break recall rises from 0.490 to 0.549 while defect-free recall stays at 0.977. Because the recalibration is label-agnostic, this geometric improvement is obtained without reweighting the objective, so it does not over-correct. The parity of SE and the failure of ECA/CA further indicate that the effect requires joint channel+spatial re-normalization of the feature map, consistent with the idea that complex defect textures need a richer re-normalization than channel gating alone. In this view the attention module acts as a data-free regularizer that raises the effective support of rare classes, which is exactly why it is the robust lever on scarce real defects.

5.3. Why LDAM is unstable: over-correction of aggressive reweighting

LDAM's class-balanced reweighting (DRW) assigns the fray class (only 26 training samples) roughly 28× the majority-class weight, so the training objective becomes dominated by minority-class samples and the model sacrifices the majority class; defect-free recall collapses from 0.983 to 0.653, and net macro G-mean drops by 8.6pp. Table 16 makes the over-correction quantitative: the margin Δ_c and the class-balanced weight w_c both grow sharply as the sample count shrinks, so the two components compound rather than compensate. This is the classic failure of "aggressive reweighting over-corrects on small-sample real data": when a rare class's sample count cannot support the large weight it is assigned, reweighting introduces a new bias instead of correcting the old one. The margin component is less harmful but still negative (0.7135, −3.8pp), because on a small dataset a wider boundary for scarce classes is purchased at the cost of the majority class.

5.4. Why stacking LDAM on CBAM dilutes the gain

The two modules do not compose additively. CBAM first sharpens feature discriminability and raises the costly classes (break 0.490 to 0.549); stacking LDAM's margin and DRW on top then re-dominates the objective toward the minority classes, pulling break recall back to 0.471 and defect-free recall down to 0.838. At the default margin coefficient ($C = 0.5$) the combination reaches 0.744, below both CBAM (0.782) and CE (0.752); the C-sweep (Section 4.6) further shows this combination is highly sensitive to C and does not robustly beat CBAM at any value. This is the concrete mechanism behind the paper's central finding: *feature attention and loss reweighting act on the same long-tailed*

Table 16
LDAM margin $\Delta_c = C/n_c^{1/4}$ (C=0.5) and class-balanced weight w_c (β=0.9999) for the six MTSD classes. The rarest class receives ≈28× the majority-class weight and ≈2.3× the margin, so the two components compound.

Class (train n)	Margin Δ_c	Weight w_c
Defect-free (761)	0.095	0.0014
Blowhole (92)	0.161	0.0109
Uneven (82)	0.166	0.0122
Break (68)	0.174	0.0148
Crack (46)	0.192	0.0218
Fray (26)	0.221	0.0385

problem through different channels, and on small real data the loss channel over-corrects rather than complements the feature channel.

5.5. Why CS-CBAM collapses: train/inference inconsistency

CS-CBAM, which injects the class prior $\lambda \log(N/n_c)$ into attention logits, collapses to G-mean 0, predicting almost everything as the majority class. The failure mechanism is train/inference inconsistency: during training the label is known, so the attention logit receives the bias and tail-class samples get larger attention weights, and the classifier learns the shortcut "large attention response = tail class"; at inference the label is unknown, the bias cannot be injected, and the shortcut fails. This collapse is a robust mechanistic conclusion: *injecting a label-dependent signal into a label-agnostic operator creates an unbridgeable distribution shift between training and inference.* It is consistent with (Kang et al., 2020), which localizes the long-tail problem to the classifier level.

5.6. The correct placement of the cost-sensitive signal

Together, the four mechanisms above yield a practically meaningful rule for inspection-system design: *on small real long-tailed datasets the cost-sensitive signal should be realized in the feature layer through label-agnostic attention, not in the loss layer through reweighting or margin.* Feature attention protects the rare, costly classes by sharpening their discriminative features without re-dominating the objective; loss-layer cost calibration over-corrects in every form tested, and injecting the cost signal into attention necessarily collapses. This rule is the main transferable takeaway of the paper for practitioners who build long-tailed industrial inspection systems.

5.7. Comparison with existing magnetic tile methods

On the MTSD dataset, existing methods mostly report average accuracy: CRAB (Li et al., 2024) reaches 99.50%, LPAR-Net (Lv et al., 2024) reaches 93.63%, and FFCNN (Xie et al., 2021) also achieves high accuracy. Our focus differs: under long tails and cost asymmetry, average accuracy hides the misses of rare defects: the CE baseline already reaches 0.905 average accuracy, yet its hardest break recall is only 0.490. We therefore use macro G-mean as the primary metric; CBAM lifts it from 0.752 to 0.782 and markedly improves the recall of the costly break defect. Because training/test splits and metric conventions differ across methods, the literature numbers in Table 2 are for reference only; our core conclusion (the correct placement of the cost-sensitive signal) comes from the controlled ablations in Section 4.6.

5.8. Answering the research questions

The three research questions posed in Section 1 now have concrete, evidence-backed answers. *RQ1* (*which module helps?*): feature-layer attention helps and loss-layer cost calibration does not: CBAM raises macro G-mean by +3.1pp with the lowest variance, whereas every LDAM variant lowers it (−3.8pp to −8.6pp). *RQ2* (*where should the cost signal act?*): on this small real dataset neither the feature path nor the training objective is the robust home of the cost signal; the feature path should be reserved for label-agnostic attention, and the cost signal is best deferred to the decision rule (thresholding against the enterprise's $C_{\text{miss}}/C_{\text{re}}$ ratio), which is applied after training and requires no retraining. *RQ3* (*do the modules compose?*): they do not compose additively; stacking LDAM on CBAM dilutes the gain from 0.782 to 0.744, so the two modules should be treated as alternatives in a decoupled design rather than as cumulative upgrades. These answers, and the design rule that follows from them, are summarized in Table 18.

6. Applications

The central finding (that label-agnostic feature attention, not loss-layer reweighting, is the robust lever on long-tailed defect data) translates directly into practice. This section describes how CSAR-Net is deployed on a magnetic tile inspection line, quantifies the economic benefit of its recall gains under quality-cost theory, and distills a transferable design recipe for other long-tailed industrial surfaces.

6.1. Deployment in an industrial inspection line

CSAR-Net can be dropped into an existing inspection line with almost no architectural cost. A magnetic tile line typically images each tile under controlled illumination, runs a defect classifier, and routes the suspicious tiles to re-inspection. Because the backbone is the standard ResNet18 and the two CBAM blocks add only 0.7% parameters (11.18M to 11.26M) with negligible FLOPs on the small 14×14 and 7×7 maps, the model runs at the same latency as the plain backbone and can be served by an edge GPU or an industrial PC. The LDAM loss and the CS-CBAM bias term add no parameters and vanish at inference, so deployment requires no change to the network itself, only to the evaluation metric and the decision rule. Fig. 8 sketches the pipeline: image acquisition, preprocessing (resize to 224×224, grayscale replicated to three channels), CSAR-Net inference, a cost-aware decision rule, and routing to accept or re-inspect. This is the same flow as a conventional accuracy-optimized classifier; the difference lies entirely in how the classifier is trained and evaluated.

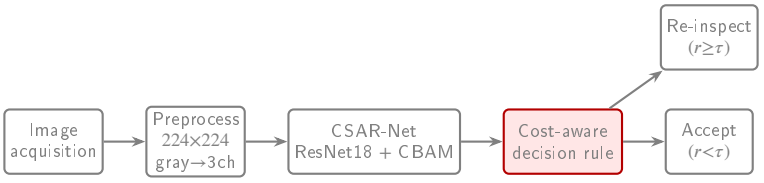


Figure 8: Deployment pipeline of CSAR-Net on a magnetic tile inspection line. Only the classifier training, the evaluation metric, and the decision rule differ from a conventional accuracy-optimized system; the inference-time footprint is identical to the plain ResNet18 backbone.

6.2. Cost–benefit analysis under quality-cost theory

The recall gains are economically meaningful because miss costs dominate. Under quality-cost theory (Juran & Gryna, 1993), a miss (a defective tile passed as normal) is an *external failure cost* (the most expensive category), whereas a false alarm is only an internal re-inspection cost. Let π_c be the class prior, $\mathrm{FNR}_c = 1 - \mathrm{Recall}_c$ the miss rate of class c, and FPR_c the rate at which tiles of other classes are falsely routed to re-inspection. The expected cost per inspected tile is

$$\mathbb{E}[\mathrm{cost}] = \sum_{c=1}^{K} \pi_c \Big[\mathrm{FNR}_c \, C_{\mathrm{miss}}^{(c)} + \mathrm{FPR}_c \, C_{\mathrm{re}} \Big], \tag{10}$$

where $C_{\mathrm{miss}}^{(c)}$ is the external-failure cost of missing class c and C_{re} is the much smaller re-inspection cost of a false alarm. Because safety-critical misses satisfy $C_{\mathrm{miss}}^{(c)} \gg C_{\mathrm{re}}$, the first term dominates, and a method that raises the recall of a costly class while holding false alarms roughly fixed yields an economic gain even when average accuracy is unchanged.

On the test split (≈269 tiles) this is concrete rather than hypothetical. Table 17 computes the expected miss and false-alarm counts from the 3-seed recall means and the per-class test counts. The CE baseline misses ≈22.3 defect tiles and falsely flags ≈3.2 defect-free tiles; CBAM misses ≈19.0 defect tiles and falsely flags ≈4.4. In other words, CBAM saves about 3.4 defect misses at the price of about 1.1 extra false alarms per ≈269 tiles; since $C_{\mathrm{miss}} \gg C_{\mathrm{re}}$, the net saving $\approx 3.4\, C_{\mathrm{miss}} - 1.1\, C_{\mathrm{re}}$ is strongly positive. In contrast, the LDAM-stacked variant that forces break recall up through aggressive reweighting simultaneously drags defect-free recall down to 0.838 (Fig. 9), flooding the re-inspection queue with ≈31 false alarms: the over-correction failure expressed in cost terms. The decision threshold can be set by trading the ROC against the enterprise's actual $C_{\mathrm{miss}}/C_{\mathrm{re}}$ ratio (Provost & Fawcett, 1997; Elkan, 2001), an instance of cost-sensitive learning applied to the *decision rule* rather than to the feature or loss layer (Elkan, 2001; He & Garcia, 2009).

Table 17
Expected miss and false-alarm counts per test set (≈269 tiles), computed from the 3-seed recall means and per-class test counts. CBAM saves ≈3.4 defect misses at the price of ≈1.1 extra false alarms.

Quantity (per ≈269 tiles)	CE	CBAM
Blowhole misses (n=23)	5.3	3.7
Break misses (n=17)	8.7	7.7
Crack misses (n=11)	2.3	2.3
Fray misses (n=6)	0.7	1.0
Uneven misses (n=21)	5.3	4.3
Total defect misses	22.3	19.0
Defect-free false alarms (n=191)	3.2	4.4

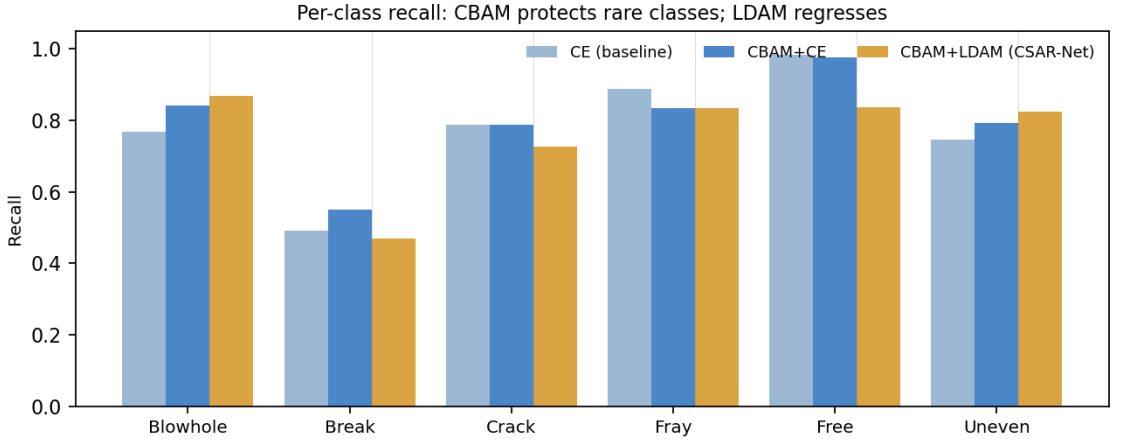


Figure 9: Per-class recall of CE, CBAM+CE, and CBAM+LDAM on MTSD (3-seed mean). CBAM lifts the rare, costly classes (break, blowhole, uneven) while keeping defect-free recall nearly intact; stacking LDAM pulls break and defect-free recall back down.

6.3. Design guidance and generalization

For practitioners building a long-tailed industrial classifier, the ablation (Fig. 10) condenses into an actionable recipe: (i) add a label-agnostic channel+spatial attention block to the deep residual stages; this is the single robust gain and requires no hyperparameter beyond the standard CBAM; (ii) keep the objective as plain cross-entropy, or a small margin at most, and avoid aggressive class-balanced reweighting, which over-corrects on scarce real defects (He & Garcia, 2009; Drummond & Holte, 2003); (iii) evaluate with macro G-mean, never average accuracy. The recipe transfers across backbones and domains: the same CBAM blocks attach to a larger ResNet or a lightweight MobileNet (Howard et al., 2017; Sandler et al., 2018) for different throughput budgets, and the imbalance-aware evaluation is expected to carry over to other long-tailed industrial surfaces: steel (NEU-DET) (Song & Yan, 2013), metallic parts (Tao et al., 2018), and deployment-oriented surface-defect systems (Tabernik et al., 2020; Roth et al., 2022), and industrial anomaly-detection benchmarks (Bergmann et al., 2019). Detection-based pipelines (from YOLO (Redmon et al., 2016; Bochkovskiy et al., 2020) to segmentation networks (Ronneberger et al., 2015)) and alternative backbones (deeper ResNets (He et al., 2016), EfficientNet (Tan & Le, 2019), and vision transformers (Dosovitskiy et al., 2021; Liu et al., 2021)) share the same long-tailed label structure, so the feature-layer attention rule should transfer. Resampling remedies such as SMOTE (Chawla et al., 2002) remain complementary when a scarce class has enough structure to synthesize, and recent surveys (Bhatt et al., 2021) catalog the wider surface-defect landscape. Calibrating the exact cost ratio with enterprise quality-cost data, and few-shot adaptation to novel defect types, are the natural next steps for deployment.

6.4. Limitations, deployment risks, and reproducibility

A number of caveats bound the transfer of these results to a production line. First, the study is single-dataset: every conclusion is drawn on MTSD, and cross-domain transfer to steel (NEU-DET), metal parts, or PCB is argued

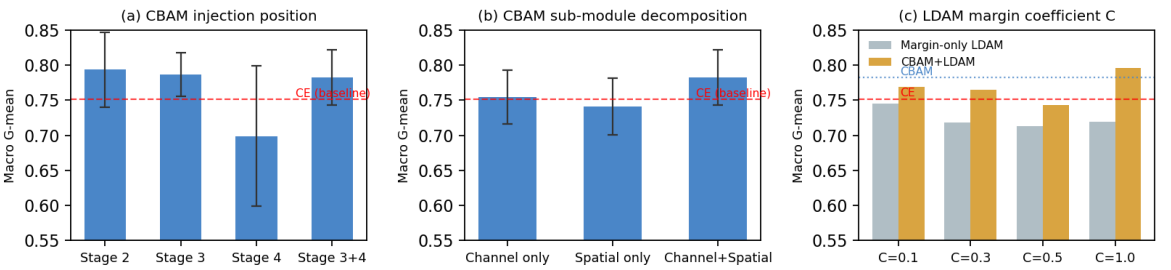


Figure 10: Ablation summary: (a) injection position is robust except for stage 4 alone; (b) neither channel-only nor spatial-only clears the baseline; (c) the CBAM+LDAM combination is non-monotonic and fragile in the margin coefficient C. Feature-layer attention is the only robust lever.

by analogy rather than by direct experiment, so the magnitude of the CBAM gain on other surfaces remains to be measured. Second, seed-to-seed variance is non-trivial (macro G-mean standard deviations of 0.03–0.05); the +3.1pp CBAM gain is robust relative to this noise, but the finer ordering among the top three injection positions is not. Third, the cost asymmetry is proxied by the inverse class frequency rather than by an enterprise's measured quality-cost ledger, so a deployment should replace this proxy with an actual $C_{\text{miss}}/C_{\text{re}}$ ratio. Fourth, the classifier is closed-world: novel defect types absent from training are necessarily misclassified, so an open-set or anomaly-detection wrapper (Bergmann et al., 2019; Roth et al., 2022) is advisable. On the reproducibility side the pipeline is deliberately simple and fully deterministic: it is implemented in PyTorch (Paszke et al., 2019), the ResNet18 backbone is trained from scratch on MTSD (grayscale replicated to three 224×224 channels) without ImageNet pretraining, and the model is optimized with Adam (Kingma & Ba, 2015) (learning rate 10^{-3}, weight decay 10^{-4}, cosine annealing, 40 epochs, batch size 32) over three seeds; fixed cuDNN settings remove algorithm nondeterminism, and every ablation reuses the identical recipe. ImageNet pretraining (Krizhevsky et al., 2012) and multi-surface validation are the most immediate extensions for even scarcer domains.

7. Conclusion

This paper addresses long-tailed, cost-asymmetric magnetic tile surface defect recognition and proposes a Cost-Sensitive Attention Residual Network, CSAR-Net, assembled from three decoupled modules: a ResNet18 backbone for hierarchical feature extraction, CBAM for label-agnostic feature-layer attention, and LDAM for label-dependent loss-layer cost calibration. On MTSD, feature-layer attention is the robust lever: CBAM raises macro G-mean from 0.752 to 0.782 (+3.1pp, the lowest variance) and lifts the recall of the hardest, costliest break defect from 0.490 to 0.549. A complete ablation localizes the source of the gain and the cause of the degradation: feature attention is the only stable gain, every form of loss-layer cost calibration over-corrects (-3.8pp to -8.6pp), stacking LDAM on CBAM dilutes the attention gain to -0.8pp, and injecting the cost signal into attention (CS-CBAM) necessarily collapses. These findings translate into a clear design rule for long-tailed inspection systems: align the metric with miss costs via macro G-mean, and protect the rare, costly classes through feature-layer attention rather than through loss-layer reweighting.

Table 18
Summary of findings and their design implications. Each finding maps to a concrete, transferable rule for long-tailed inspection systems.

Finding	Evidence	Design implication
Feature attention is the robust gain	CBAM +3.1pp, lowest variance	Add label-agnostic attention to deep stages
Loss-layer calibration over-corrects	LDAM -8.6 / margin -3.8 / reweight -4.8pp	Keep cross-entropy (or a small margin)
Stacking LDAM on CBAM dilutes the gain	CSAR-Net 0.744 vs. CBAM 0.782	Do not stack calibration onto attention
Injecting cost into attention collapses	CS-CBAM G-mean $\to 0$ in all seeds	Keep attention label-agnostic; cost at decision rule

The limitations are as follows: the experiments are on a single magnetic tile dataset, and per-class recall varies across the 3 seeds; generalization to other industrial surfaces (steel, PCB, castings) remains to be verified; and the cost asymmetry is currently proxied by class frequency rather than an enterprise's actual quality-cost data. Future work will extend to more industrial defect datasets, explore stronger backbones and minority-class generation, and calibrate precise cost structures with enterprise quality-cost data in real production lines.

CRediT authorship contribution statement

[First author]: Conceptualization, Methodology, Supervision, Funding acquisition, Writing - Review & Editing. Chenyang Xie: Methodology, Software, Investigation, Validation, Formal analysis, Writing - Original Draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The source code, training configuration, and figure-generation scripts that reproduce the experiments reported in this paper are openly available at `https://github.com/otreeevol/csar_net`. The Magnetic Tile Surface Defects (MTSD) dataset is a public benchmark introduced by Huang et al. (2020) and was used as released; no new data were generated in this study. Trained model checkpoints are not distributed in the public repository because of their size, and are available from the corresponding author on reasonable request.

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